AC circuits

"DC power supply"

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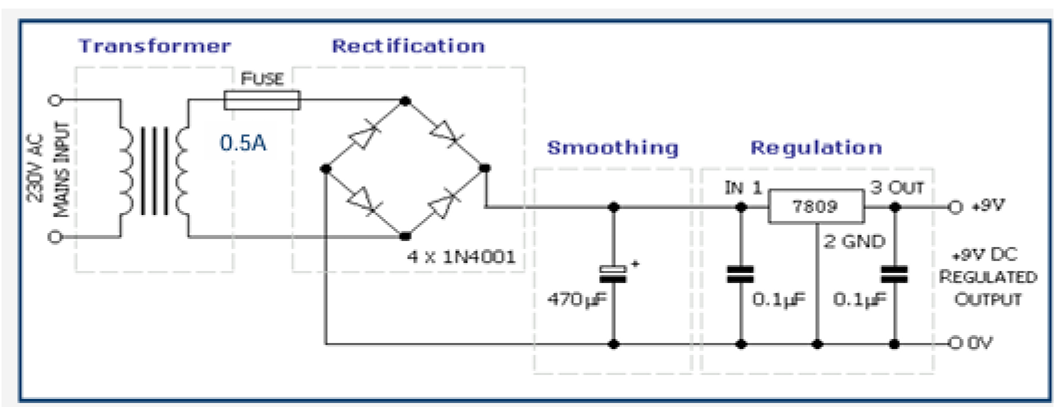
Introduction

A DC power supply is an essential electronic system used to convert electrical energy from an AC source into a stable DC output. In many electronic devices, DC voltage is required to operate circuits safely and efficiently. Since the mains electricity supply provides a high-voltage alternating current, it must be converted and regulated before it can be used in electronic applications.

The main objective of this project is to design and implement a DC power supply that converts the 220V AC mains voltage into a regulated DC voltage. This is achieved through different stages including voltage transformation, rectification, and voltage regulation. The final output provides a stable and safe DC voltage suitable for powering electronic circuits and components

Idea of operation



The DC power supply operates by converting the 220V AC mains voltage into a regulated 9V DC output through several sequential stages.

First, the AC mains voltage is applied to a step-down transformer. The transformer reduces the high 220V AC input to a lower AC voltage suitable for electronic circuits. A fuse is connected in series with the input to protect the circuit from overcurrent and short-circuit conditions.

Next, the reduced AC voltage is fed into a bridge rectifier composed of four 1N4001 diodes. This stage converts the alternating current (AC) into pulsating direct current (DC) by allowing current to flow in only one direction during both halves of the AC cycle.

After rectification, a smoothing capacitor is connected across the output. This capacitor reduces voltage ripples and smooths the pulsating DC into a more stable DC voltage.

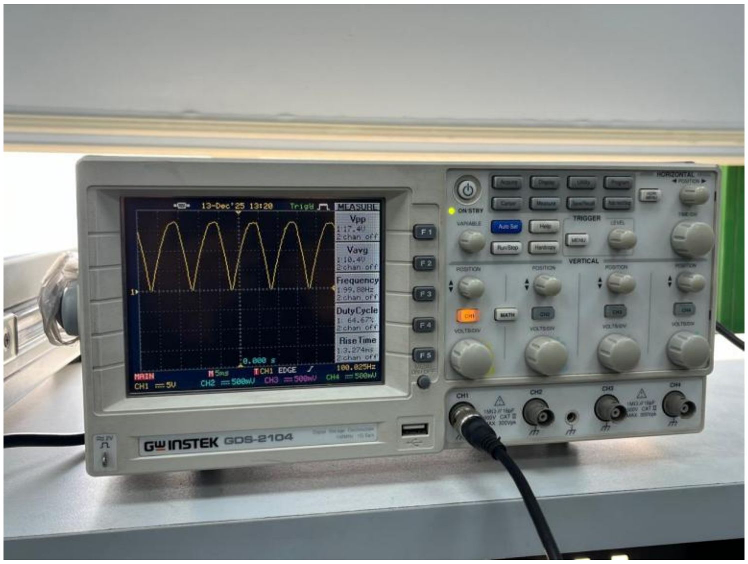
Finally, the smoothed DC voltage is applied to a voltage regulator (7809). The regulator maintains a constant 9V DC output regardless of small variations in the input voltage or load current. Additional small capacitors are used at the input and output of the regulator to improve stability and reduce noise.

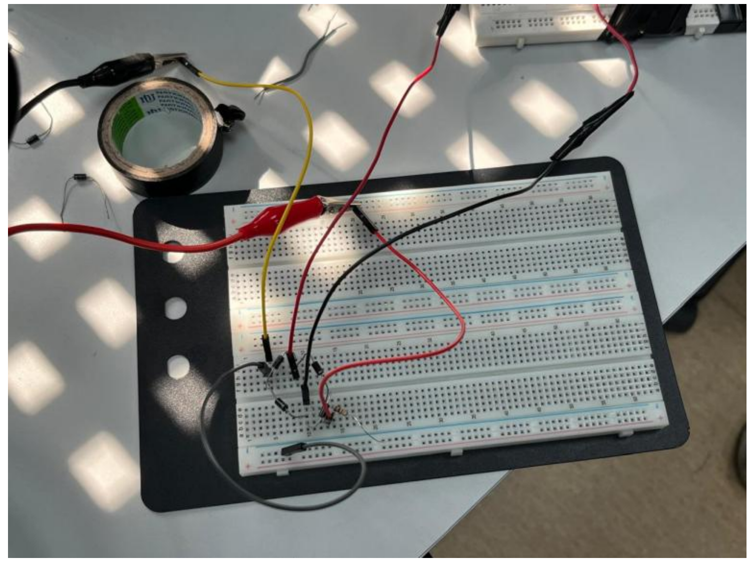
As a result, the circuit provides a stable and regulated 9V DC output suitable for powering electronic devices

Technical Specifications

- **Input Voltage:**
  220V AC, 50 Hz (Mains Supply)
- **Protection:**
  Input fuse: 0.5 A
- **Transformer:**
  Step-down transformer (220V AC to low-voltage AC)
- **Rectification:**
  Full-wave bridge rectifier
  Diodes: 4 × 1N4001
  Maximum diode current: 1 A
- **Filtering (Smoothing):**
  Electrolytic capacitor: 470 µF
  Purpose: Ripple voltage reduction
- **Voltage Regulation:**
  Voltage regulator IC: 7805

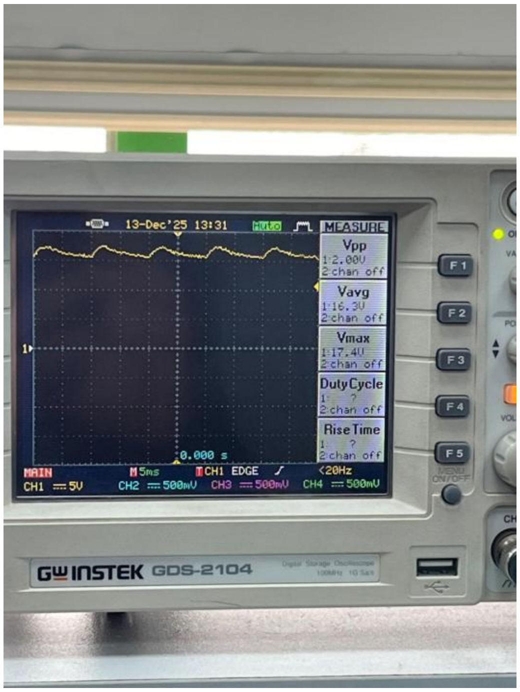
Full wave rectifier

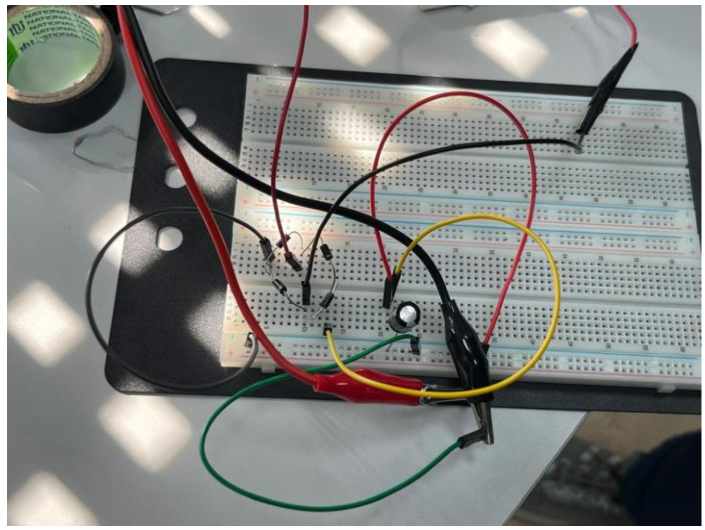




The rectifier stage converts the AC voltage from the transformer into a pulsating DC voltage using a bridge of diodes.

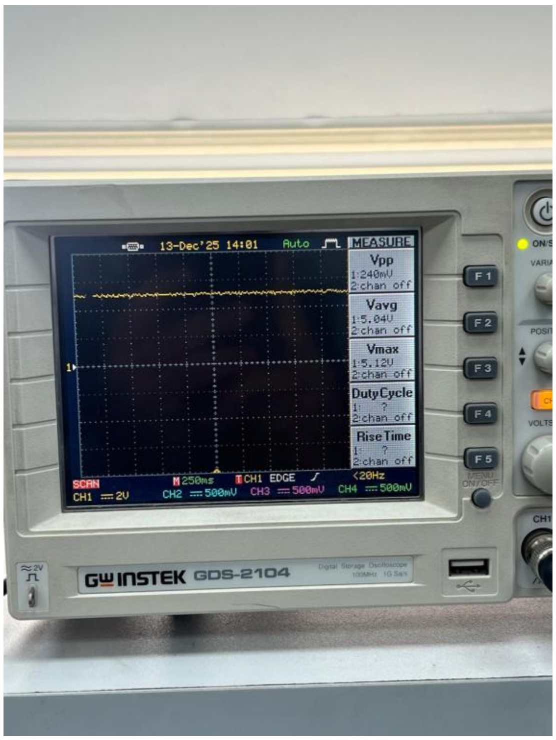
Smoothing stage

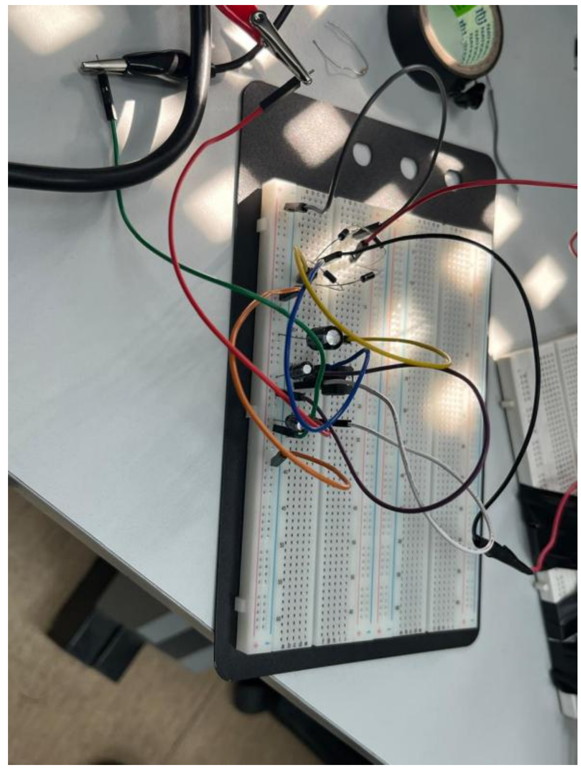




The smoothing stage uses a capacitor to reduce voltage ripples and provide a more stable DC output

Regulation stage





the regulation stage maintains a constant DC output voltage by using a voltage regulator to eliminate variations caused by input or load changes.